6 **(a)** A sound wave travels through air. Fig. 6.1 shows a pressure–time graph for the air at one place.

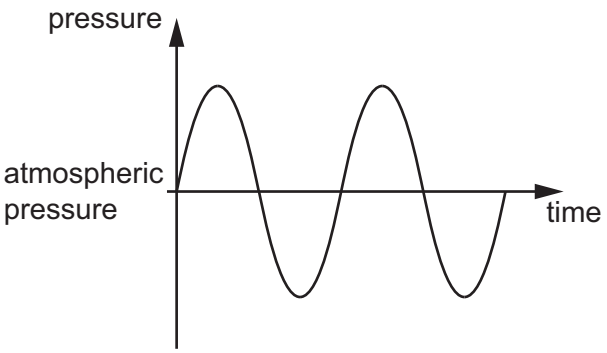

Fig. 6.1

(i) On Fig. 6.1:

- label **one** point C to indicate a compression
- label **one** point R to indicate a rarefaction.

[2]

(ii) Explain why this graph **cannot** be used to find the wavelength of the sound wave.

.....

..... [1]

(iii) The sound becomes louder and of lower pitch.

State what happens to:

the amplitude of the sound

.....

the frequency of the sound.

.....

[1]

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(b) A sound of frequency 13 kHz is transmitted through water.
The speed of sound in water is 1500 m / s.

Calculate the wavelength of this sound in water.

wavelength = ... [3]

(c) State the approximate speed of sound in air.

speed = ... [1]

[Total: 8]

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